# X-Y Positioner Driven By Electrostatic Motor†

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**Abstract** **An X-Y positioner driven by electrostatic motors name dual excitation electrostatic motor was developed. Unique configuration of the positioner utilizing a characteristic of the motor balances inertia mass in the two axes to achieve good dynamic characteristics of the positioner. A prototype positioner was fabricated and tested. The prototype generated a thrust force of approximately 35N in each axis, and a positioning resolution of 200μm was obtained. To improve the resolution, electrode structure of the motor was modified, and very smooth motion was obtained.**

## I. Introduction

Recently, need for very small electromechanical systems, known as micromachines, becomes strong. For such systems, electrostatic motors are considered to be preferable to electromagnetic motors because of its ease of fabrication and high force per volume or weight ratio, As a result, many electrostatic motors have been developed[1..3]. In such situation, a possibility of applying electrostatic motors to devices with larger dimensions up to some centimeters is pointed out. To realize such idea, the authors are developing several types of such electrostatic motors[4..6]. Among these motors, dual excitation multiphase electrostatic drive (DEMED)[6] had the best performance in terms of power per weight ratio. Additionally, the performance of the motor was even superior to typical electromagnetic motors. From these results it follows that the electrostatic motor is applicable to not only micro-sized devices but also normal sized ones such as VCRs and computer peripherals allowing their miniuarization. As one of such non-micro applications, this paper introduces an X-Y positioner driven by the electrostatic motor, fabricates a prototype and tests it.

## II. Dual Excitation Multiphase Electrostatic Drive

As depicted in Fig. 1, *dual excitation multiphase electrostatic drive* (DEMED) consists of a pair of thin plastic sheets called "slider" and "stator." In both films, multiphase parallel electrodes are embedded. When defined multiphase voltages are applied to the electrodes in the two films, the slider film runs upon the surface of the stator film perpendicularly to longitudinal axes of the electrodes. There is a choice of profile of the voltages and electrode structure, and formerly, two types of DEMED that were driven by rectangular pulse and sinusoidal ac have been developed. Fig. 2 is a schematic cross sectional view of the DEMED which can be driven by 3-phase sinusoidal ac voltages. In case of this type of DEMED, application of 3-

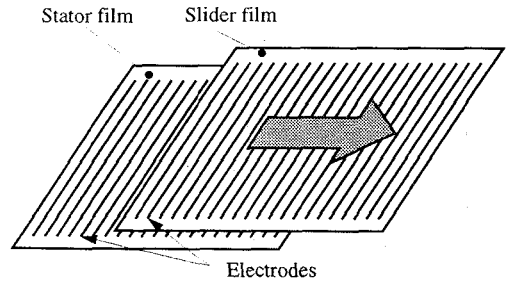



Fig. 1 Basic structure of dual excitation multiphase electrostatic drive

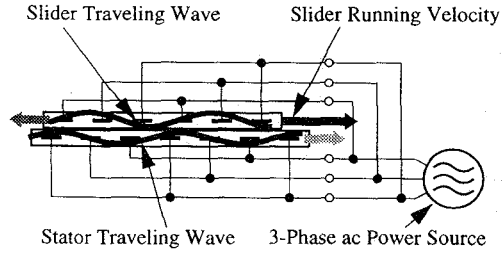



Fig. 2 Schematic cross sectional view of DEMED which can be driven by a 3-phase ac voltage source. The 3 phases of the voltages are connected to the electrodes of the slider and the stator in orders reversed to each other. Two bold lines in the stator and slider indicate potential distributions generated by the excitation.

† This work was done mainly at Kanagawa Academy of Science and Technology.

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phase ac voltage to the 3-phase electrodes both in the slider and the stator causes two potential distributions as displayed by bold lines in this figure, and electrostatic force generated between the two distributions propels the slider film. The motor is a kind of synchronous motor, whose operation speed is proportional to the frequency of applied ac.

### III. X-Y Positioner Using DEMED Technology

Usually, to obtain X-Y motion, we build two sets of one-axis linear tables, and place one of them upon the moving element of the other arranging their moving direction perpendicularly to each other. However, such structure increases inertia mass of the lower axis to degrade dynamic characteristics of the positioning device.

In case of DEMED, the electrostatic force acts only perpendicularly to the electrodes' longitudinal axes; the slider is free in the longitudinal axis as shown in Fig. 3. This follows that, if we arrange two sets of the motors perpendicularly to each other and connect their sliders as shown in Fig. 4, we can obtain X-Y motion by controlling the two motors separately.

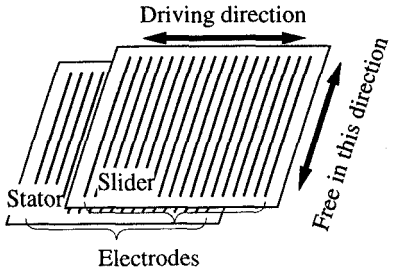



Fig. 3 Direction in which electrostatic force acts. Electrostatic force acts only in lateral direction to the electrodes, shown by bold allow, and slider can be freely moved in the longitudinal direction.

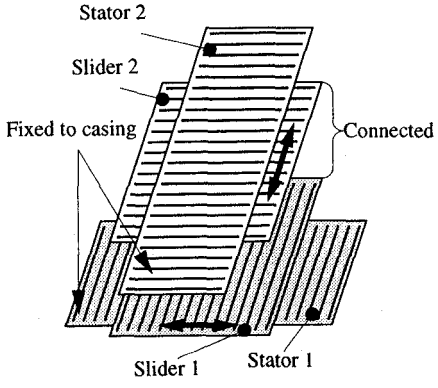



Fig. 4 Basic concept of X-Y positioner. Two motors are arranged perpendicularly to each other and connected at their slider. Stators of both motors are fixed.

### IV. Fabrication

A prototype X-Y positioner which was driven by two sets of ac-drive DEMEDs was built. The prototype consisted of two DEMED motors which shared completely the same design. Fig. 5 shows structure of one axis of the motors. The motor had 6 sliders and 6 stators stacked alternately in order to generate multiplied thrust force of single slider-stator pair as shown in Fig. 5. Fig. 6 displays cross sectional view of slider and stator film. The films were processed as flexible printed board. In the films, electrodes were embedded at a regular interval of 200μm, and each electrode had a width of about 100μm. Using through-hole technique, the electrodes were connected into 3 phases in each film. Between the films, a small number of glass balls with a diameter of 20μm were sandwiched so as to reduce the friction between the films and keep the gap between them. As a result of using the glass balls, friction coefficient between a slider and a stator decreased down to 0.05. As shown in Fig. 7, the electrodes were processed over areas of 80x100mm and 120x60mm in the slider and a stator respectively. Because of these sizes of the electrode areas, this motor could generate a movement with a range of 20mm, and its slider could be freely moved also in a range of 20mm.

As shown in Fig. 8, the two sets of the electrostatic motors were arranged perpendicularly to each other and connected at their sliding elements. In order to restrict rotation of the sliding elements, they were placed on a linear guide that were placed on another linear guide arranged perpendicularly to it. The stating elements of both motors were fixed to a plastic casing. The casing was filled with dielectric fluid in order to prevent breakdown in gap between a slider and a stator.

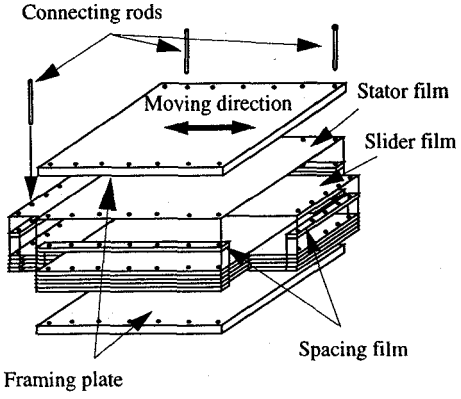



Fig. 5 Structure of one axis of motor. This motor consists of 6 sliders and stators stacked alternately.

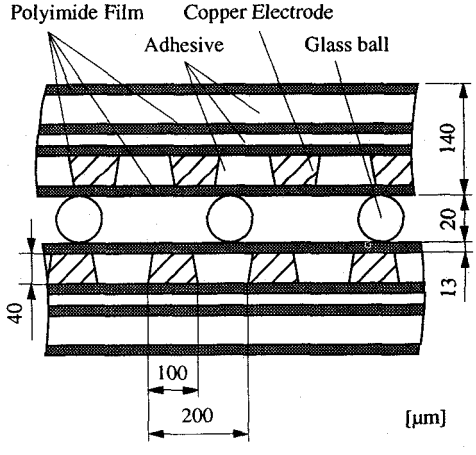



Fig. 6 Cross sectional view of slider and stator stack



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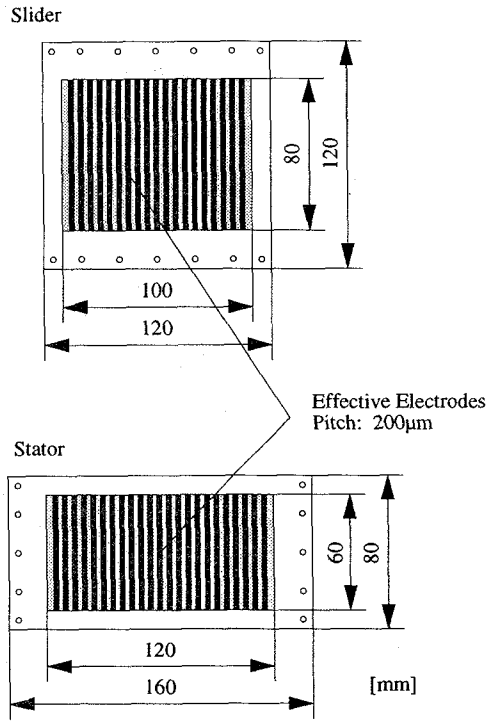



Fig. 7 Layout of electrodes on a slider and a stator

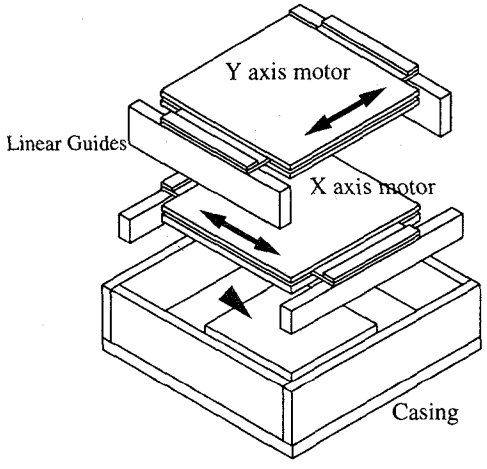



Fig. 8 Structure of the prototype X-Y positioner

As a result of these processes, the prototype X-Y positioning device weighed 4.2kg in total. Moving element has inertia masses of 860g and 1500g in X direction and Y direction respectively. The difference of inertia in the directions was caused by unbalanced X-Y guide structure. Fig. 9 and Tab. I show dimensions and distribution of mass of the prototype respectively.

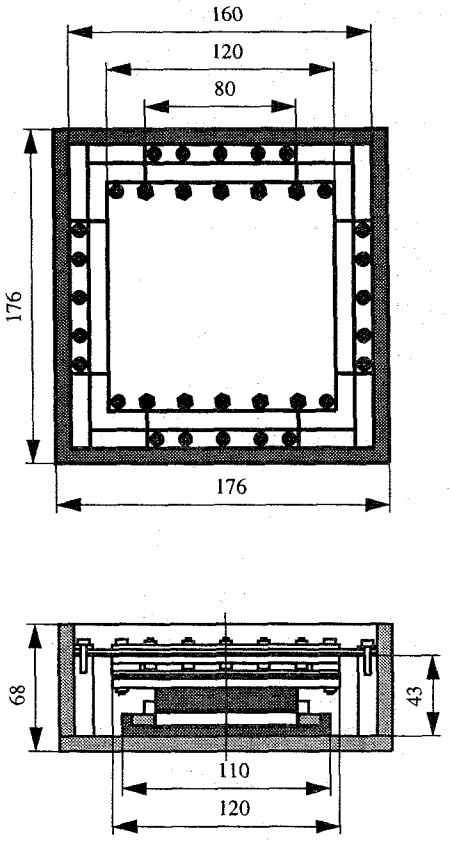



Fig. 9 Dimensions of the prototype

Tab. I Distribution of mass of the prototype

| Element | Part | Mass [g] |
|---|---|---|
| Sliding Element | Slider Films | 32 |
| | Framing | 321 |
| | Others | 49 |
| | SUBTOTAL | 402 |
| Stating Element | Stator Films | 38 |
| | Framing | 278 |
| | Others | 29 |
| | SUBTOTAL | 346 |
| Linear Guides | Sliding Element (X) | 456 |
| | Stating Element (X)+Sliding Element (Y) | 705 |
| | Stating Element (Y) | 387 |
| | Others | 42 |
| | SUBTOTAL | 1590 |
| Casing | Panels | 862 |
| | Others | 40 |
| | SUBTOTAL | 902 |
| Dielectric Liquid | | 1000 |
| TOTAL | | 4240 |

## V. Experimental Results

Fig. 10 is relationship between the thrust forces in the two axes and excitation voltage to the electrodes. The thrust force was measured by simple weight lift-up experiments. The thrust force was approximately proportional to square of the voltage and reached the top values at the voltage which caused breakdown between two neighboring electrodes in a slider or a stator. The breakdown voltages of the films were varied from $1400V_{rms}$ to $2400V_{rms}$, but in the most films, they were around $1800V_{rms}$. As top values of thrust forces, 35N and 36N were obtained at this voltage in the X and Y axes respectively.

Fig. 11 and 12 show a movement of the moving element of the prototype X-Y positioner when a right triangle with a height



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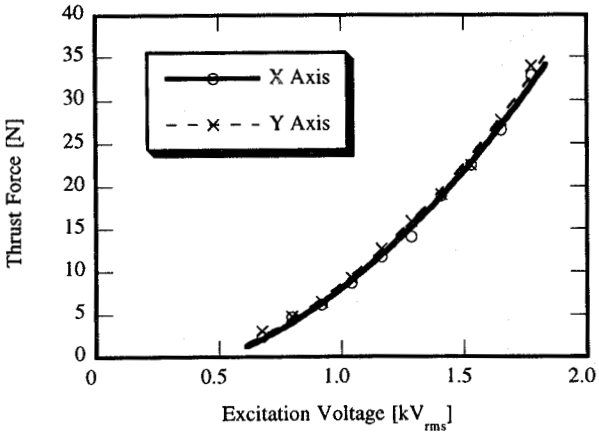



Fig. 10 Relationship between thrust forces in two axes and voltage of applied ac

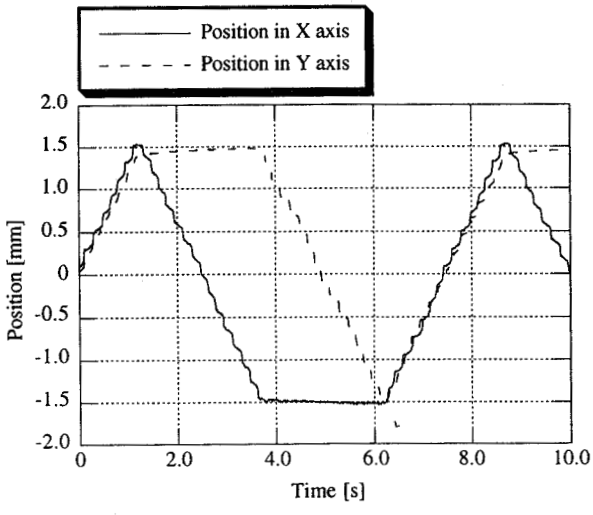



Fig. 11 Motion of the moving element of the prototype positioner

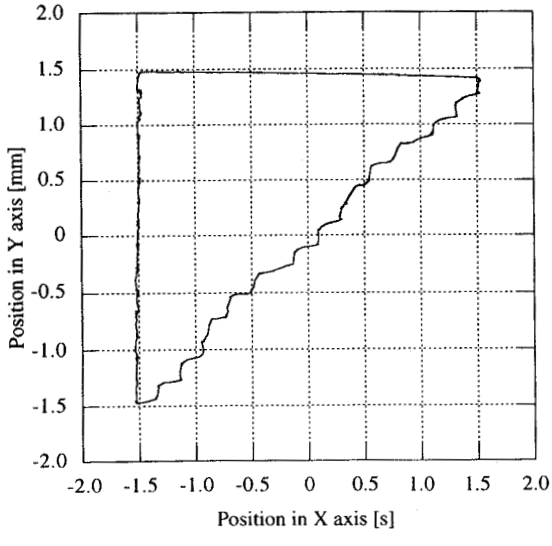



Fig. 12 Trajectory of the moving element when a triangle was commanded

of 3mm and a width of 3mm was commanded. Stepwise motion found in Fig. 11 was caused by rippled thrust force of the motor. Since the electrodes in the slider and the stator were located at an interval of 200μm, step length of the movement became the same value of 200μm. This means that positioning resolution of this device cannot be smaller than 200μm. Fig. 13 shows a trajectory of the moving element when a circle with 10mm diameter was commanded. The path was also fluctuating because of 200μm of resolution.

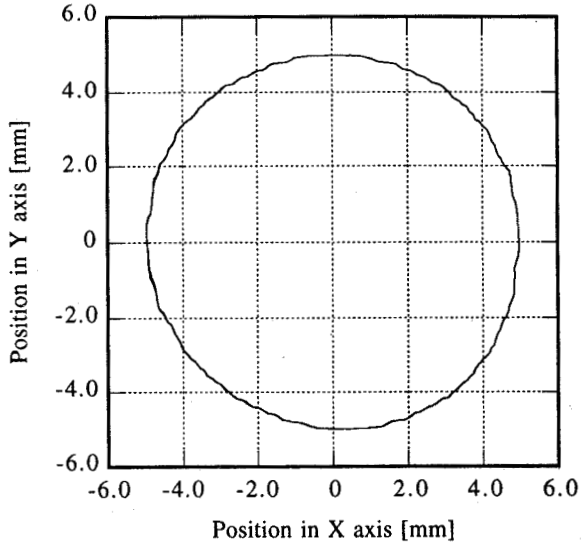



Fig. 13 Trajectory of the moving element when a circle was commanded

As mentioned before, a slider is free from electrostatic force in the direction perpendicular to longitudinal axes of the electrodes. However, in reality, attractive electrostatic force generated between sliders and stators caused friction between the films, and it could constrain the smooth motion of the slider. Fig. 14 is relationship between the smallest force that causes slip of the sliding element in the direction perpendicular to its operating one as shown in Fig. 15. The friction force in the direction was also proportional to the applied voltage to the electrodes. This shows that electrostatic force was dominating the friction.

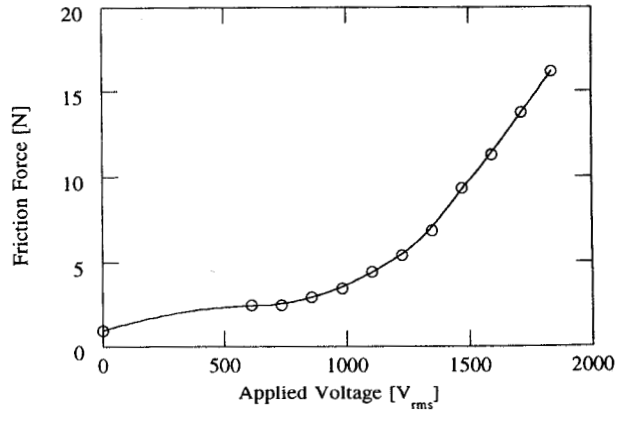



Fig. 14 Relationship between friction force and applied voltage



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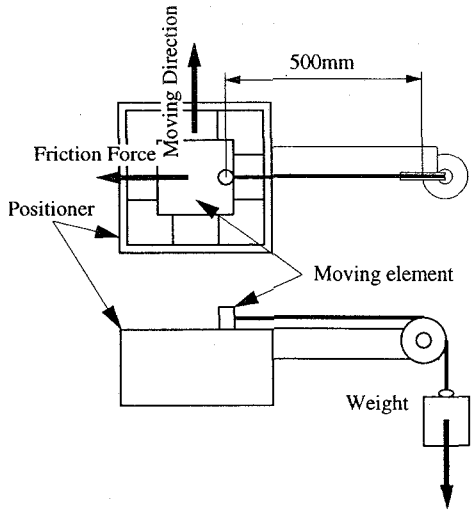



Fig. 15 Measurement of friction force in lateral direction The smallest weight that can causes slip of sliding element of a motor determines the friction force.

Use of linear guides to prevent the rotation of the moving element of the X-Y positioner degrades compactness and dynamic characteristics of the positioner. To examine if it is possible to eliminate this problem, a prototype X-Y positioner without linear guides was tested. The prototype with its moving element put on a plain nylon plastic board instead of linear guides worked very well if a torque loaded on the moving element was smaller than a certain value. Fig. 16 shows relationship between the smallest torque that can rotate the moving element and excitation voltage for operation.

In the case of this prototype, positioning resolution was defined by the interval at which electrodes were located. To improve this performance, a motor which has 3-phase electrodes and 4-phase one in its stator and slider respectively as shown in Fig. 17 was fabricated and tested. To operate a DEMED, the wave lengths of potential distributions generated in its slider and stator must be the same; interval of periodic phases must be the same in a slider and a stator. Therefore, if electrodes which have different number of phases are processed in a slider and stator, ratio of the regular intervals of the electrodes in the two films must be inverse ratio between the numbers of phases.

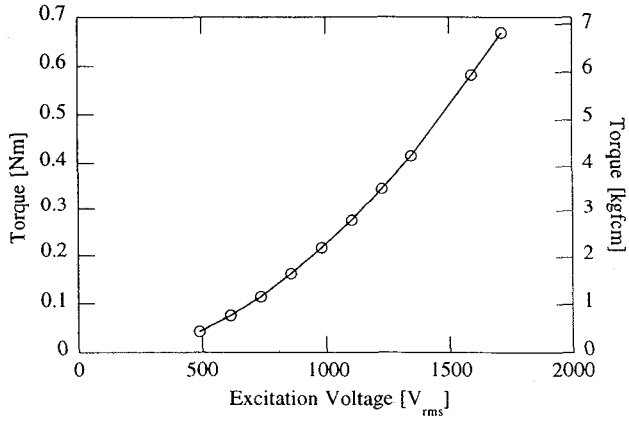



Fig. 16 Relationship between the smallest torque that can rotate sliding element and applied voltage.

For example, if the electrodes are connected into 4 phases and 3 phases in the slider and the stator respectively, ratio of regular interval of the electrodes is 3:4. In the case of tested motor, intervals of electrodes in its slider and stator were 150μm and 200μm respectively. Fig. 18 shows movement of the sliding element of the motor. Very smooth motion of sliding element was obtained by modifying the number of phases.

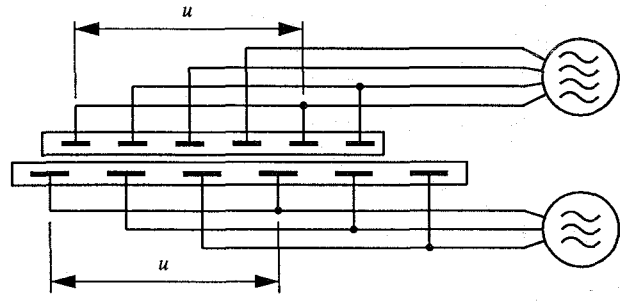



Fig. 17 Schematic cross sectional view of a DEMED which has 4-phase electrodes and 3-phase ones in its slider film and stator film respectively. Interval of periodic phases , indicated by *u*, are the same in the slider and the stator.

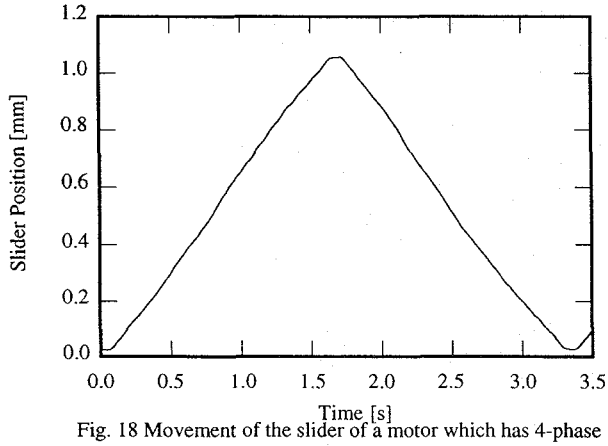



Fig. 18 Movement of the slider of a motor which has 4-phase electrodes and 3-phase ones in its slider and stator respectively.

## VI. Discussions

Total mass of the prototype is very large in comparison with that for film. This result spoils merits of DEMED as a light-weight and high-power motor. The reasons lying behind this degradation seem mainly

- mass of linear guides
- mass of casing and framing
- mass of dielectric liquid in dead volume

Among these masses, that for linear guides can be eliminated if a torque loaded on the moving element is smaller than a certain value.

In this prototype, design of casing and framing was not optimized for decrease of the their volume or mass. By refining the design of these parts, their mass can be reduced, but it seems difficult to decrease that less than 50%.

By quitting use of linear guides, dead volume existing in the lower half of the casing can be eliminated (see Fig. 9). As a result, mass of the dielectric fluid can be reduced approximately



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to the half of original one, but it seems also difficult to decrease it much more.

Since the thrust force of stacked DEMED is proportional to the number of layers, thrust force of an X-Y positioner consisting of single-layer motors can be estimated as 6N in each axis. This force seems quite enough for some applications such as optical stages. Employing such configuration reduces required strength of framing and casing of the positioner, and as a result, it becomes possible to built a very thin, compact and light-weight positioner.

Positioning resolution of the tested prototype, 200μm, was very poor in comparison with typical commercially available ones such as optical stages positioned by screws and nuts. However, the possibility to improve the resolution by modifying numbers of phases of the electrodes was given in this paper. In the authors' former work, it was confirmed that resolution length of a motor which had the same electrode structure can be, at least, as small as one third of the interval of the 4-phase electrodes when simple pulse voltages to its electrodes[5] were applied. Additionally, experimental result shown in Fig. 18 follows that resolution of the motor can be improved even more using a technique similar to "micro step." This technique is typically used for electromagnetic stepping motors to improve their resolution.

## VI. Conclusions

An X-Y positioner driven by two sets of dual excitation multiphase electrostatic drives were introduced. A unique configuration of this positioner, in which the two motors are connected at their sliding element, equalized inertia mass in the X and Y directions realizing balanced dynamic characteristics in the two directions. A prototype consisting of two sets of 6-layer motors was fabricated and successfully operated. The prototype weighed 4.2kg and generated a thrust force of approximately 35N in each axis, and a positioning resolution of 200μm was obtained.

The moving element of the positioner was fixed on linear guides to prevent its rotation at first, but by later experiment, it was confirmed that the positioner could be operated without the guides under a certain torque loaded on the moving element.

The mass of the positioner spoiled the merit of dual excitation multiphase drive as light-weight and high-power motor, but there is possibility of reducing the mass by removing the linear guides, refining the design of the framing and casing, and reducing dead volume in the casing.

As long as the number of electrodes phases are the same in a slider and a stator, the positioning resolution could not be smaller than the regular interval of the electrodes. However, use of different numbers of phases in the slider and the stator improved the resolution, and very smooth slider motion was obtained. This result showed the possibility of "micro step" technology applied to electrostatic motors as well as to typical electromagnetic stepping motors.

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